

Disproof of Conjecture 00000000156

TLMC submission

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Conjecture 00000000156. *The probability that the determinant of a random ± 1 $n \times n$ matrix is prime in absolute value is $\sim c/n$ (with an explicit formula for c).*

Verdict: FALSE.

1 The attack

Proposition 1. *Let A be an $n \times n$ matrix with all entries in $\{-1, 1\}$. Then $2^{n-1} \mid \det A$.*

Proof. Subtract the first row of A from each of the rows $2, \dots, n$. This row operation does not change the determinant. In the resulting matrix A' , every entry of rows $2, \dots, n$ lies in $\{0, -2\}$. Multiplying each of those $n - 1$ rows by $1/2$ multiplies the determinant by 2^{n-1} , so

$$\det A = 2^{n-1} \det B$$

for an integer matrix B (its first row is the first row of A , and its remaining rows have entries in $\{0, -1\}$). Hence $2^{n-1} \mid \det A$. $\square$

Corollary 1. *For $n \geq 3$, no ± 1 $n \times n$ matrix has prime $|\det|$.*

Proof. If $\det A = 0$, $|\det A|$ is not prime. Otherwise, by Proposition 1 and $n \geq 3$, the integer $|\det A|$ is a positive multiple of $2^{n-1} \geq 4$; in particular it is even and at least 4. The only even prime is 2, and $4 \nmid 2$, so $|\det A|$ is composite. $\square$

Corollary 2 (Disproof). *Let p_n be the probability that $|\det A|$ is prime for a uniformly random ± 1 $n \times n$ matrix. Then $p_n = 0$ for every $n = 1$ and every $n \geq 3$ (Corollary 1; for $n = 1$, $|\det| = 1$). Since $p_n \equiv 0$ for all $n \geq 3$, we have $p_n/(c/n) \rightarrow 0 \neq 1$ for every constant $c > 0$. The conjectured asymptotic $\sim c/n$ (with any positive explicit c) is false.*

2 Boundary: exact small- n behavior

- $n = 1$. $\det \in \{-1, 1\}$: never prime, $p_1 = 0$.
- $n = 2$. $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$ with $ad, bc \in \{-1, 1\}$, so $\det A \in \{-2, 0, 2\}$ and $|\det A| = 2$ is prime exactly when $ad \neq bc$, which happens in 8 of the $2^4 = 16$ matrices:

$$p_2 = \frac{8}{16} = \frac{1}{2}.$$

This is the *only* dimension in which a prime determinant can occur (Corollary 1). Matching $p_2 = c/2 = 1/2$ would force $c = 1$, but then $c/n = 1/n$ versus the true values $p_n = 0$ for all $n \geq 3$; the conjecture fails there and hence asymptotically.

- $n = 3$ (**exhaustive**, $2^9 = 512$ **matrices**). All 512 determinants are divisible by 4, and $\det \in \{-4, 0, 4\}$: no primes, $p_3 = 0$.
- $n = 4$ (**exhaustive**, $2^{16} = 65536$ **matrices**). All 65536 determinants are divisible by 8, and $\det \in \{0, \pm 8, \pm 16\}$: no primes, $p_4 = 0$.
- $n = 5, 6, 7, 8$ (**randomized**). 3000 samples each (reproduce.py, seed 156): 100% divisible by 2^{n-1} , zero prime $|\det|$.

3 Formal verification (core Lean, no Mathlib)

The package contains a Lean 4 (toolchain v4.33.1, core only) development:

- `no_prime_det`: the arithmetic heart — a number m with $4 \mid m$ (written $m = 2 \cdot (2 \cdot q)$) is never prime; applied to $m = |\det A|$ for $n \geq 3$, where $2^{n-1} \mid \det A$ gives $4 \mid |\det A|$, this yields $p_n = 0$ for all $n \geq 3$ and refutes the $\sim c/n$ asymptotic;
- `two_is_possible`: $m = 2$ is prime and even, pinpointing $n = 2$ as the only feasible case;
- `exhaustive decide` checks over the 16 determinants of all 2×2 and the 512 determinants of all $3 \times 3 \pm 1$ matrices (bit enumeration, regenerated by `reproduce.py`): divisibility by 2^{n-1} , absence of prime $|\det|$, and the count 8 of prime determinants at $n = 2$.

`Check.lean` runs `#print axioms` on every theorem; all report “*does not depend on any axioms*”. There are no `sorry` statements.

4 Reproduction

```
python3 reproduce.py          # exhaustive n <= 4, randomized n = 5..8
cd lean4 && lake build && lake env lean Check.lean
```